Project 3: Demand Estimation Lab

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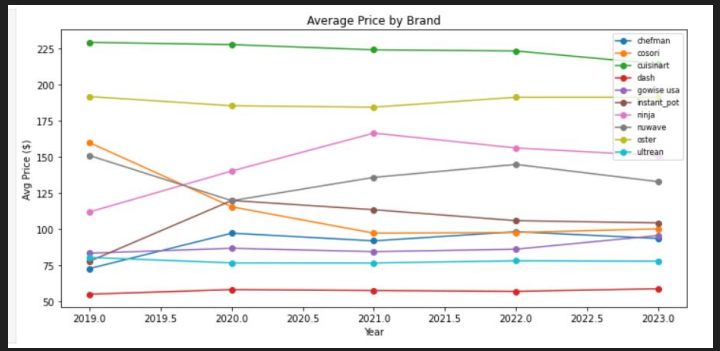
Description of the Air-Fryer Market

The air fryer market grew significantly between 2019 and 2023 with 10 major brands competing for consumer purchases on Amazon.

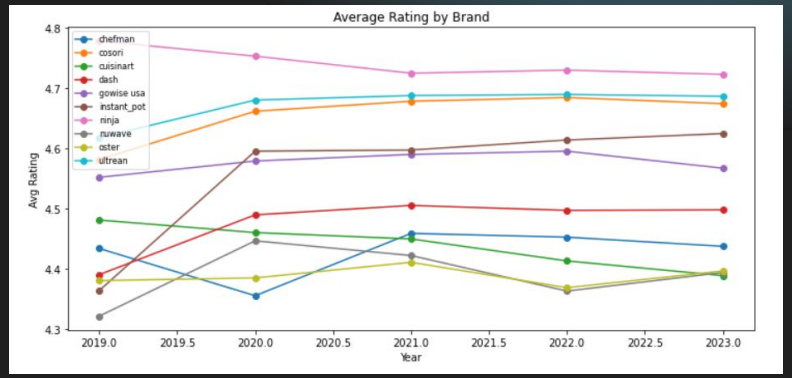
- Ninja is the dominant brand holding the largest average market share at 19.2% and growing steadily over the 5-year period
- Instant pot made the most dramatic rise going from just 1.6% share in 2019 all the way to 22.9% in 2023 making it one of the fastest growing brands in the market
- Gowise USA experienced the steepest decline falling from the market leader position at 29.2% share in 2019 down to just 5.3% in 2023.
- Cuisinart is the most expensive brand at an average price of $224 while Dash is the most affordable at just $57 on average.

Most brands sell compact style air fryers which is by far the most common product type in the market. Ratings are generally high across all brands ranging from 4.39 to 4.74 out of 5 showing, that consumers are largely satisfied regardless of which brand they choose.

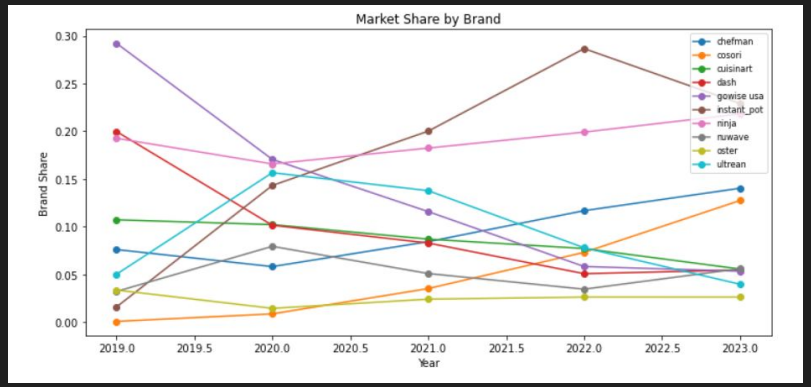
Plots showing prices, ratings, and market shares over time



Average Price by Brand



Average Rating by Brand



Market Share by Brand

Explanation of the Demand Model

- To understand what drives consumers to choose one air fryer over another, I built a demand model using a method called Multinomial Logistic Regression. A demand model is a tool that helps us understand what makes consumers choose one product over another, and how sensitive they are to changes in price or product quality. Simply put, the model looks at each brand's market share in each year and tries to explain why some brands have bigger shares than others based on price, ratings, and product features. Instead of working with market share directly, I used the log market share as our outcome which is a standard math transformation that makes the model work correctly. I also added brand and year controls so the model accounts for the fact that some brands are just generally more popular and that overall demand shifts from year to year. The model was estimated using simple linear regression on 50 observations covering 10 brands over 5 years and it explained about 76.3% of the variation in market share across brands and years. Overall, the model gives us a reliable way to measure how price and product features affect consumer demand in the air-fryer market.

Estimated Price Coefficient (and its sign)

- The model estimated a price coefficient of -0.0377 which means that as a brand's price goes up its market share goes down. The **negative sign** is exactly what we expect because in real life when something gets more expensive fewer people buy it. If the coefficient were positive it would mean higher prices lead to more purchases which makes no sense and would break all of our cost and profit calculations. This one number is the foundation of everything in our analysis, because it tells us how sensitive consumers are to price changes in the air fryer market.

Discussion of the most important product features

- **Compact style** is the most important and statistically significant product feature in the model with a coefficient of 9.82. This means brands that sell more compact air fryers tend to have significantly higher market shares than brands that do not. Almost every brand in the market sells compact fryers which shows that this is what consumers want the most.

- **Window Style** has the second largest positive coefficient at 12.88 meaning it is also associated with higher demand though its effect is just borderline significant. Chefman is the only brand that offers window style products in a meaningful share of its lineup making it somewhat unique in the market. This feature could be a growing opportunity for brands looking to differentiate themselves.

- **Dual Basket + Rotisserie** are actually associated with lower demand based on their negative coefficients of -9.51 and -5.67 in the model. However, these effects are not statistically significant so we cannot say with full confidence that they hurt demand. What we can say is that these features are very rare in the market and consumers do not appear to be seeking them out.

Estimated Unit Costs, Markups, & Share-Weighted Profits

- **Unit Costs** vary widely across brands from just $27.94 for Dash all the way up to $194.90 for Cuisinart showing that brands are operating at very different cost structures. Ninja and Nuwave also have high unit costs at $112.49 and $109.05 reflecting that their products are more expensive to produce. Cheaper brands like Ultrean and Gowise USA have much lower unit costs which allows them to price competitively in the market.

- **Markups** are surprisingly similar across all 10 brands ranging narrowly from $27.23 for Oster to $32.85 for Ninja. This means that no single brand is dramatically overcharging or undercutting the competition in terms of of profit margin on each unit sold. The tight range of markups suggests that competition in the air-fryer market keeps pricing power roughly equal across brands.

- Ninja has the highest **Share-Weighted Profit** at 6.31 followed closely by Instant Pot at 6.01, making these two brands the clear winners in the market. Gowise USA is third at 4.61 despite losing a lot of market share over the years, showing it maintained solid profitability. Oster has the lowest share weighted profit at just 0.68, meaning it generates very little profit relative to its presence in the market.

Conclusion about the Brands' relative positions in the market

- **Ninja** and **Instant Pot** are the clear leaders in the air-fryer market with the highest market shares, the strongest demand, and the highest share weighted profits, making them the two brands any competitor should be most worried about. **Cuisinart** and **Oster** are the most expensive brands but they have been losing ground over time and generate relatively low share-weighted profits, suggesting that high prices alone do not guarantee success in this market. **Gowise USA** is an interesting case because it was the market leader in 2019 with 29.2% share, but collapsed to just 5.3% by 2023, showing how quickly a brand can lose its position if it fails to keep up with consumer preferences. **Cosori** is the brand to watch going forward as it grew from nearly 0% share in 2019, to 12.7% in 2023, showing the fastest growth trajectory of any brand in the market.